Name	Subject Class	Class	Candidate Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2014

BIOLOGY **9648/02**
25 AUGUST 2014
HIGHER 2 **2 hours**

Paper 2 Core Paper

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST
Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A
Answer **all** questions.

Section B
Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.
Fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8 or 9	
Total	**100**

This question paper consists of **18** printed pages.

[Turn over

1 **(a)** Fig. 1.1 below is an electron micrograph of part of a human liver cell.

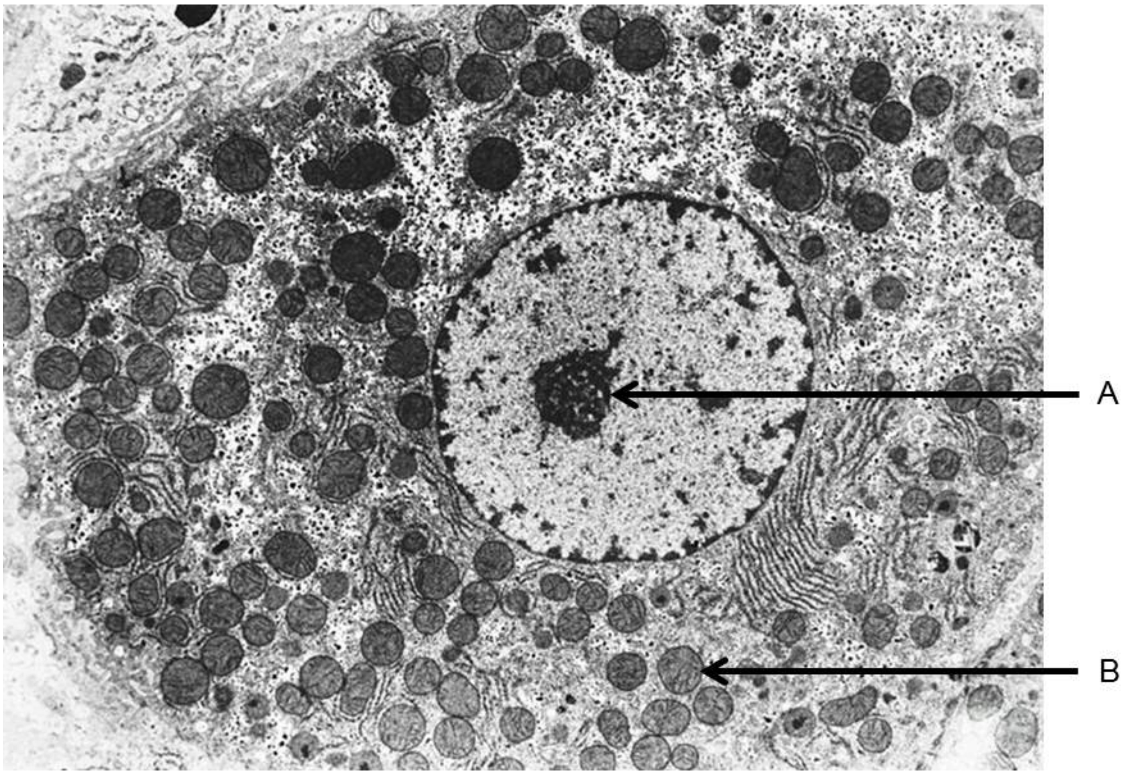

Fig. 1.1

(i) With reference to Fig. 1.1, describe the structure of **A** and state **one** difference in the roles of structures **A** and **B** in the cell.

[2]

(ii) Suggest why there is a large number of **B** in the liver cell.

[1]

For examiner's use

(iii) A large number of storage polysaccharide is found throughout the liver cell. Explain how the structure of this storage polysaccharide makes it a good energy storage molecule.

[4]

(b) It is common for patients suffering from liver disease to have symptoms such as jaundice, abdominal pain, swelling and itchy skin. To reduce the pain caused by the disease, doctors usually prescribe painkillers to the patients. A common type of painkiller is Acetaminophen and it works by blocking calcium ion channels at synapses between neurones.

(i) Explain why there is a need for calcium ion channels to transport calcium ions across the membrane.

[2]

(ii) Explain how the blocking of calcium ion channels can result in the reduction of pain in patients with liver disease.

[3]

[Total: 12]

2 (a) In 1882, the German botanist T.W. Engelmann performed an ingenious experiment to investigate the effects of different wavelengths of light on the rate of photosynthesis for a suspension of alga *Spirogyra* (a filamentous microorganism containing long, spiral chloroplasts).

In his experiment, a prism was placed between the light source and the alga filament to produce and scatter all colours of the light across the alga filament evenly. Then, aerobic bacteria were added to the alga filament suspension. All the other variables were kept constant.

After exposure to light for a certain period of time, the aerobic bacteria were found to move towards and accumulate at specific lengths along the alga filament as shown in Fig. 2.1. Engelmann concluded that the product released during photosynthesis must have caused the directional movement of aerobic bacteria.

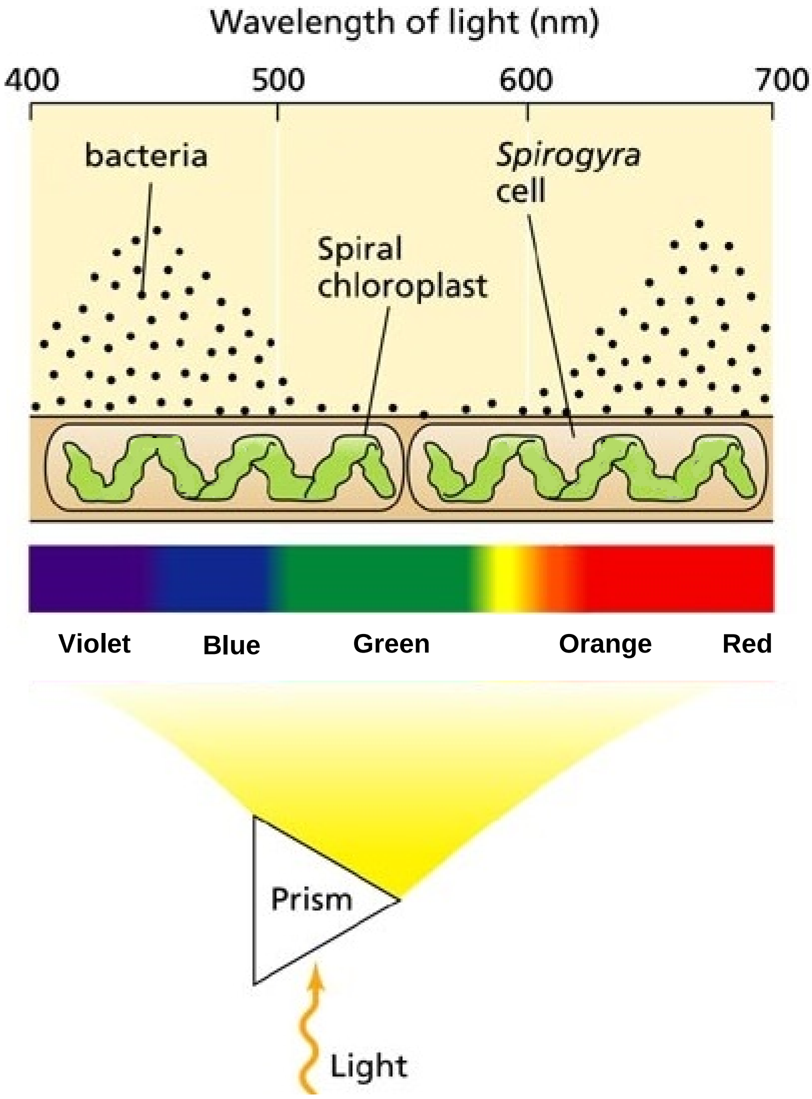

Fig. 2.1

With reference to the information above and Fig. 2.1,

(i) suggest how the rate of photosynthesis was determined in this experiment.

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.... [1]

For examiner's use

(ii) explain why the rate of photosynthesis can be determined as suggested in part (i).

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.... [1]

(iii) describe the relationship between the rate of photosynthesis and the wavelengths of light.

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.... [2]

(iv) account for the rate of photosynthesis observed for the alga filament exposed to the wavelength of light at 550 nm.

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.... [3]

For examiner's use

(b) The directional movement of cells towards a specific chemical stimulus is known as chemotaxis. It plays critical roles in regulating physiological processes such as the migration of white blood cells to sites of infection and metastasis of cancer cells. The migration of cells involves the rearrangement of a type of cytoskeleton known as actin filament.

Fig. 2.2 illustrates a model of the chemical signalling involving G-protein coupled receptor (GPCR) and different cellular proteins (Dock, Elmo, Rac), which occurs during the migration of most eukaryotic cells.

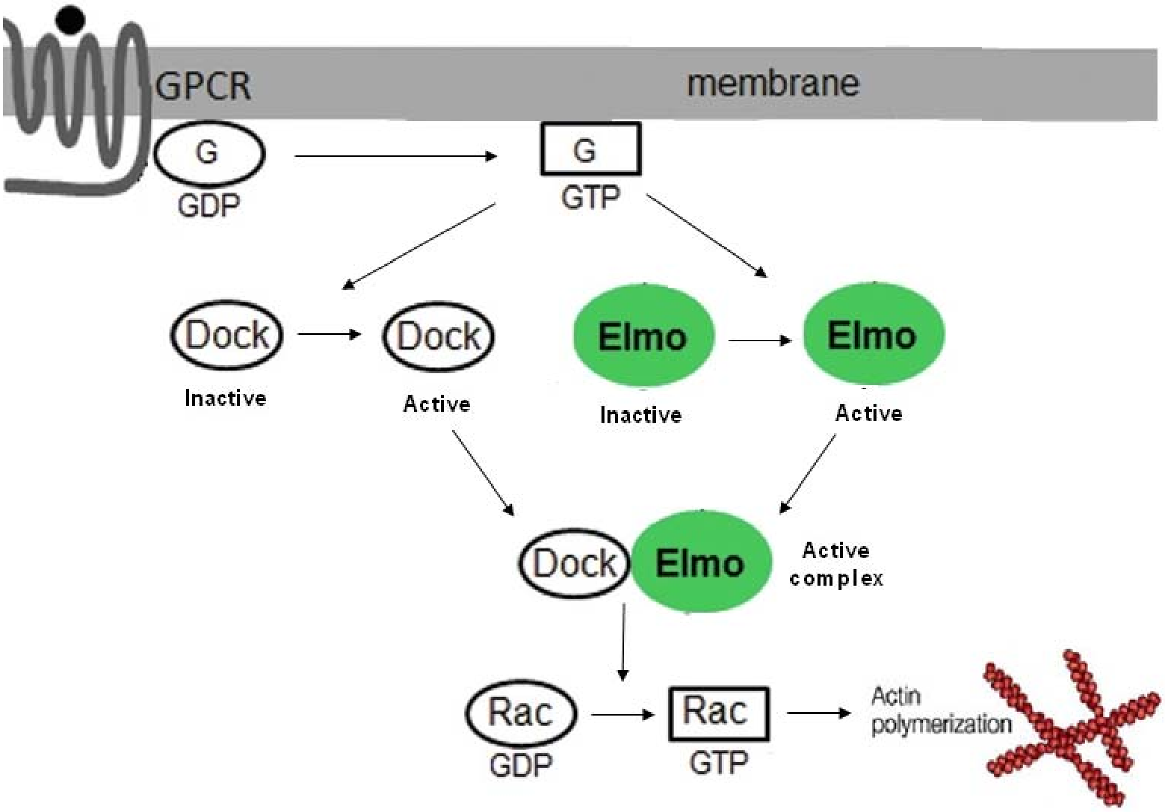

Fig. 2.2

With reference to Fig. 2.2, describe how a chemical signal molecule bound to the GPCR brings about the cellular response.

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.... [3]

[Total: 10]

For examiner's use

3 Fig. 3.1 shows the *arg* operon in *E. coli*, which is a repressible operon with three genes, *arg B, arg C* and *arg H*, that code for enzymes responsible for the biosynthesis of the amino acid arginine. Another gene upstream of the operon (not shown in Fig. 3.1) codes for a repressor protein R, which binds to the operator to regulate the operon. Arginine acts as a corepressor, which activates repressor *R*.

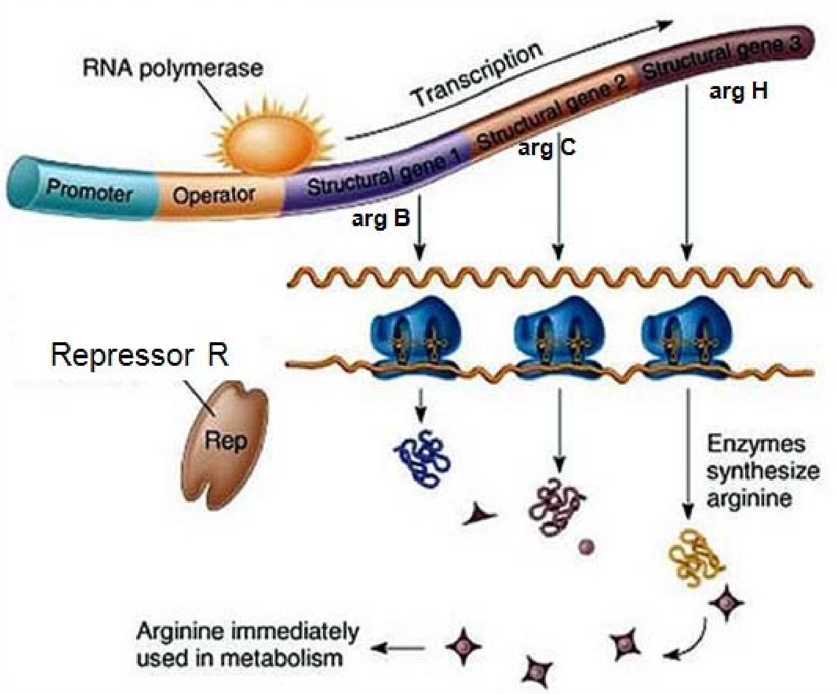

Fig. 3.1

(a) Distinguish between a structural gene and a regulatory gene in the *arg* operon.

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.... [2]

(b) Based on your understanding of a repressible system, suggest how arginine is able to control the level of enzymes involved in arginine synthesis to ensure the conservation of resources by *E. coli*.

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.... [4]

For examiner's use

(c) *E. coli* was cultured in a growth medium with all the essential nutrients needed except the amino acid arginine. Fig. 3.2 shows a graph of the concentration of enzymes that are involved in the synthesis of arginine against time. At time **X**, arginine was added to the growth medium. Using an arrow, indicate on Fig. 3.2, the time when arginine was added.

Fig. 3.2

[1]

(d) Besides having operons, prokaryotes have other means to adapt to changing environments through gene transfer.

Describe how viruses can facilitate the transfer of **many** advantageous alleles in a bacterial population.

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.... [4]

[Total: 11]

4 A group of genetics students were examining the inheritance of fruit colour and shape in a particular species of plant. When they crossed a pure breeding plant that had red, oval fruits with pure breeding plants that had yellow, long fruits, all the offspring produced were observed to have red, oval fruits.

They then proceeded to testcross two plants, **A** and **B**, from the F_2 progeny, both with red, oval fruits and the following results were obtained as shown in Table 4.1.

Table 4.1

Phenotype	Progeny of A	Progeny of B
Red, long fruits	46	4
Yellow, oval fruits	44	6
Red, oval fruits	5	43
Yellow, long fruits	5	47

(a) Based on the information and results of the crosses given above, describe the inheritance of fruit colour and shape in this species of plant.

[2]

(b) Using appropriate letters to represent the various alleles, state the genotypes of plant **A** and **B**.

Letter to represent allele for red fruit :

Letter to represent allele for yellow fruit :

Letter to represent allele for oval fruit :

Letter to represent allele for long fruit :

Genotype of plant **A**:
(with red, oval fruit)

Genotype of plant **B**:
(with red, oval fruit)

[1]

For examiner's use

(c) Draw a genetic diagram to explain the results of the testcross on plant **A** as shown in Table 4.1.

[5]

For examiner's use

(d) Using the results given in Table 4.1, calculate the relative location of the genes for fruit colour and fruit shape.

Distance between the two genes: ____________________

[1]

The group of students proceeded to measure the fruit length obtained from true breeding plants with oval fruits and those from true breeding plants with long fruits. Table 4.2 below shows the number of fruits in each fruit length category from the two respective pure breeding plants.

Table 4.2

	Fruit length / cm								
	3	4	5	6		10	11	12	13
Plants with oval fruits	4	7	6	2					
Plants with long fruits						3	6	9	5

(e) Explain the variations observed in fruit length within each group of pure breeding plants.

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____[2]

[Total: 11]

For examiner's use

5 *Xenopus* is a genus of aquatic frogs native to sub-Saharan Africa. In a developing *Xenopus* oocyte (egg cell), pre-mRNAs are processed to become mRNAs in the nucleus. They are then exported into the cytoplasm.

(a) Describe how pre-mRNA is produced in a developing *Xenopus* oocyte.

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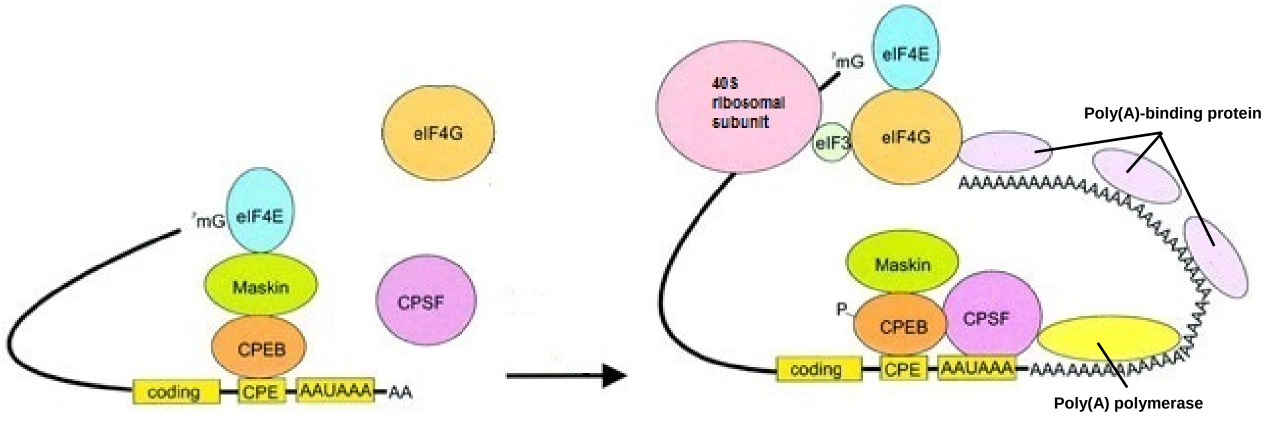
.... [4]

In immature oocytes, these mRNAs are not immediately translated but are stored in the cytoplasm for future use. Fig. 5.1a shows a translationally-repressed mRNA which has a sequence element, CPE, to which CPEB protein binds to. A repressor protein, Maskin binds to the bound CPEB. It then binds to eIF4E, a protein which binds to the modified guanosine cap (ʼmG) at the 5' end. The binding of Maskin to eIF4E prevents the binding of eIF4G protein to eIF4E, thus inhibiting the formation of the translation initiation complex. CPSF is also a protein involved in the initiation of translation.

Shortly before ovulation, these mRNAs are activated in the oocytes. In Fig. 5.1b, phosphorylation of CPEB protein causes a series of events which leads to the displacement of Maskin from eIF4E, initiating translation.



mRNA in immature oocyte | **mRNA in mature oocyte**

Fig. 5.1a | **Fig. 5.1b**

For examiner's use

(b) Contrast the structure of mRNAs in an immature oocyte to those in a mature oocyte.

[1]

(c) With reference to Fig. 5.1a and Fig 5.1b, describe how phosphorylation of CPEB protein activates translation during oocyte maturation.

[4]

(d) Describe how degradation of Maskin occurs in the cell when it is no longer needed by the cell.

[3]

(e) Suggest the advantage of control of gene expression at the level of translation over control of gene expression at the level of transcription.

[1]

[Total: 13]

For examiner's use

6 **(a)** Trinucleotide repeat expansion is a type of mutation that inserts and increases the number of base triplets in the DNA sequence. Although trinucleotide repeat expansion does not result in frameshift mutation, it can have detrimental effects on the structure and function of protein.

Fig. 6.1 illustrates an example of trinucleotide repeat expansion in the coding sequence of a gene and its effect on the polypeptide produced.

-- CAT	TCA	CAG	GTA	ATC	ATG	CTA --
His	Ser	Gln	Val	Ile	Met	Leu

↓

-- CAT	TCA	CAG	GTA	GTA	GTA	GTA	ATC	ATG	CTA --
His	Ser	Gln	Val	Val	Val	Val	Ile	Met	Leu

Fig. 6.1

(i) Describe the difference between trinucleotide repeat expansion and gene amplification in their effects on gene expression.

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.... [1]

(ii) Explain how trinucleotide repeat expansion could adversely affect the enzyme protease.

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.... [4]

(b) The huntingtin gene codes for the huntingtin (Htt) protein found in the cytoplasm. Huntington's disease is a neurodegenerative disorder caused by the increase in CAG (a base triplet coding for the amino acid glutamine) repeat in the mutated huntingtin gene.

The resultant mutant huntingtin (mHtt) protein contains an abnormally long stretch of glutamine residues, which results in the mHtt protein aggregating and accumulating within the neurones as shown in Fig. 6.2a. Fig. 6.2b shows the molecular structure of the amino acid glutamine.

Fig. 6.2a

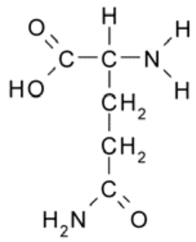

Fig. 6.2b

(i) Many globular proteins are soluble in water as the hydrophobic amino acids are buried in the interior while the hydrophilic amino acids are found on the exterior.

Based on Fig. 6.2a, Fig. 6.2b and the information above, suggest and explain why the mutant huntingtin protein molecules containing abnormally long stretch of glutamine residues aggregate and become insoluble in water.

[3]

(ii) The mutant huntingtin protein has been discovered to bind with and disrupt the function of complex II (an electron carrier) embedded in the inner membrane of mitochondria. Using your knowledge of respiration, explain how the mutant huntingtin protein will change the pH in the intermembrane space of mitochondria.

[3]

[Total: 11]

For examiner's use

7 Primate species have the potential to visually perceive three colour spectrums – blue, green and red.

Fig. 7.1 shows a phylogenetic tree of primate evolution. The common ancestor for all primate species was thought to be dichromatic, being able to perceive blue and one other colour only (hence unable to distinguish green and red colour). Only the species within the group of "apes" and "old world monkeys" are found to be consistently trichromatic and are able to perceive all three colours.

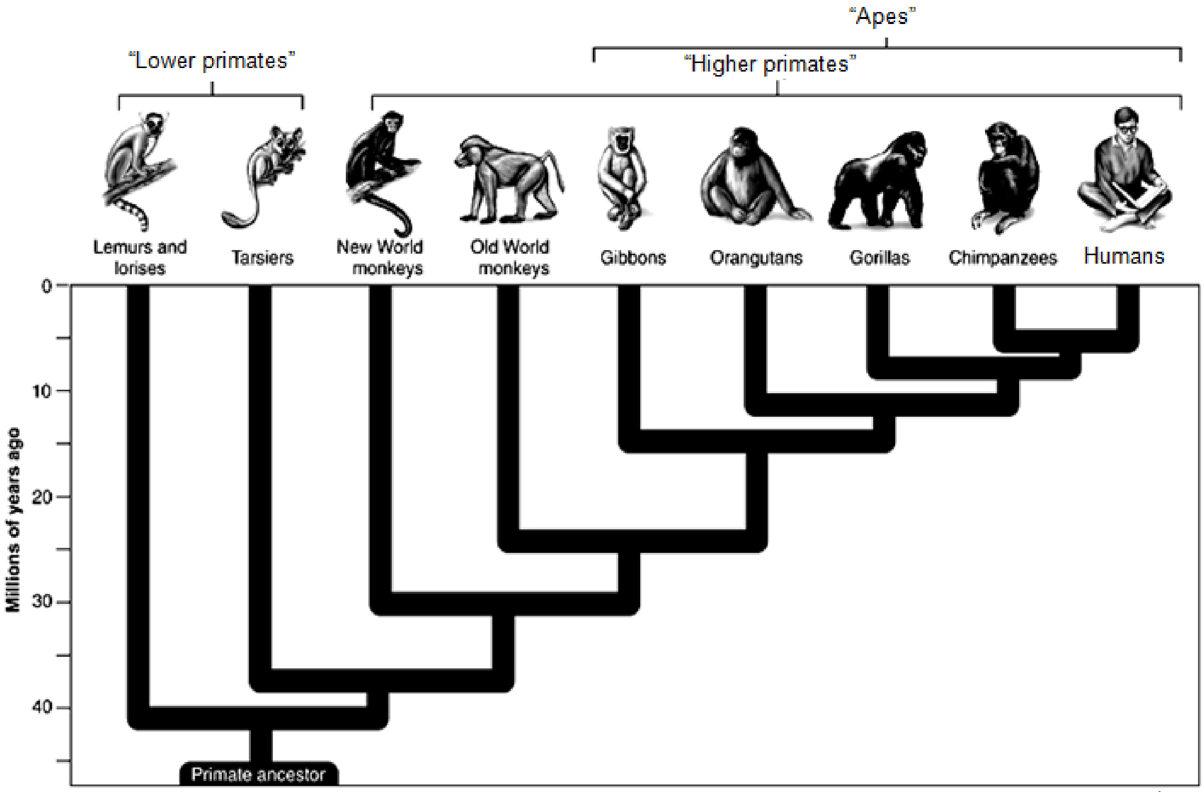

Fig. 7.1

(a) **(i)** Using an arrow, indicate on Fig. 7.1 the point where trichromatic vision is likely to have evolved in primates. [1]

(ii) Suggest how trichromatic vision may be of survival value to fruit-feeding primates in forest habitats.

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____ [1]

For examiner's use

(iii) Explain how natural selection has resulted in the evolution of trichromatic vision in a population.

[3]

Howler monkeys (*Alouatta* spp.) and night monkeys (*Aotus* spp.) are two genera belonging to the group of "new world monkeys". Both genera of monkeys are native to South and Central American forests. Howler monkeys are diurnal (active in the day) and are known for communicating by loud "howls". Night monkeys, however, are nocturnal (active at night) and communicate by owl-like "hoots".

(b) Explain how the two genera of monkeys could have evolved from a common ancestor.

[4]

All known species of "new world monkeys" have dichromatic vision, with the exception of howler monkeys which have evolved trichromatic vision.

(c) Discuss whether trichromatic vision shared by both howler monkeys and humans is an example of a homologous trait.

[3]

[Total: 12]

For examiner's use

Section B

Answer **EITHER 8 OR 9.**

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

8 **(a)** Describe how the structure of a lysosome is adapted for its function. [8]

(b) Explain the production of a small yield of ATP from anaerobic respiration in mammalian cells. [6]

(c) Contrast the products of meiosis with those of mitosis and explain how these differences come about. [6]

[20]

OR

9 **(a)** Describe the types of bonding which hold a protein in shape. [6]

(b) Explain how genotype is linked to phenotype. [8]

(c) With reference to the stages of cell signalling, describe the response of liver cells to insulin. [6]

[20]